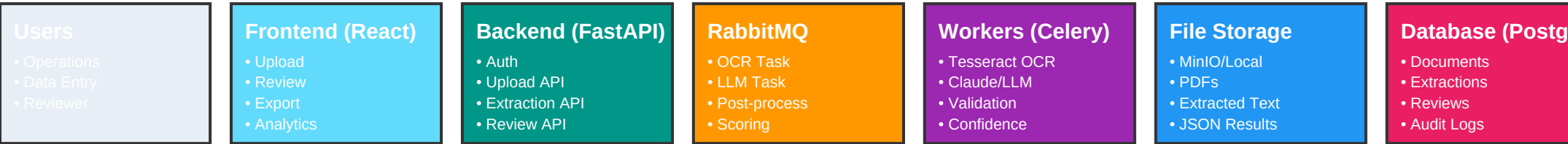


Invoice Extraction System Architecture

Open-Source Stack • Document Intelligence • Human-in-the-Loop Review



11-Step Data Processing Flow:

- | | |
|-------------------------------|------------------------|
| 1. Upload PDF (user) | 7. Score confidence |
| 2. Store file (MinIO) | 8. Route (auto/review) |
| 3. Queue task (RabbitMQ) | 9. Human review (UI) |
| 4. Extract text (OCR/PDF) | 10. Store final (DB) |
| 5. LLM extraction (Claude) | 11. Export metrics |
| 6. Validate schema (Pydantic) | |

Technology Stack (100% Open Source):

Backend: FastAPI • Frontend: React + Vue • Queue: RabbitMQ • Workers: Celery
Database: PostgreSQL • File Storage: MinIO • OCR: Tesseract • LLM: Claude API
Validation: Pydantic • Monitoring: Prometheus • Deployment: Docker + Kubernetes

Invoice Data Schema:

```
invoice_number, invoice_date, due_date, vendor_name, vendor_address
customer_name, customer_address, currency, subtotal, discount, tax, total
line_items[] { description, quantity, unit_price, amount }, confidence
```